



Vocabulary

LLM

Large Language Models (LLMs) are AI systems trained on vast text data to understand and generate human-like language. They work as prediction engines, analyzing input and predicting the next most likely token. LLMs perform tasks like text generation, translation, summarization, and Q&A. Understanding token processing is key to effective prompt engineering.

Tokens

Tokens are fundamental units of text that LLMs process, created by breaking down text into smaller components like words, subwords, or characters. Understanding tokens is crucial because models predict the next token in sequences, API costs are based on token count, and models have maximum token limits for input and output.

Context Window

Context window refers to the maximum number of tokens an LLM can process in a single interaction, including both input prompt and generated output. When exceeded, older parts are truncated. Understanding this constraint is crucial for prompt engineering—you must balance providing sufficient context with staying within token limits.

Hallucination

Hallucination in LLMs refers to generating plausible-sounding but factually incorrect or fabricated information. This occurs when models fill knowledge gaps or present uncertain information with apparent certainty. Mitigation techniques include requesting sources, asking for confidence levels, providing context, and always verifying critical information independently.

Agents

AI agents are autonomous systems that use LLMs to reason, plan, and take actions to achieve specific goals. They combine language understanding with tool usage, memory, and decision-making to perform complex, multi-step tasks. Agents can interact with external APIs and services while maintaining context across interactions.

Prompt Injection

Prompt injection is a security vulnerability where malicious users manipulate LLM inputs to override intended behavior, bypass safety measures, or extract sensitive information. Attackers embed instructions within data to make models ignore original prompts and follow malicious commands. Mitigation requires input sanitization, injection-resistant prompt design, and proper security boundaries.

Model Weight/Parameters

Model weights and parameters are the learned values that define an LLM's behavior and knowledge. Parameters are the trainable variables adjusted during training, while weights represent their final values. Understanding parameter count helps gauge model capabilities - larger models typically have more parameters and better performance but require more computational resources.

Fine-tuning vs Prompt Engineering

Fine-tuning trains models on specific data to specialize behavior, while prompt engineering achieves customization through input design without model modification. Prompt engineering is faster, cheaper, and more accessible. Fine-tuning offers deeper customization but requires significant resources and expertise.

AI vs AGI

AI (Artificial Intelligence) refers to systems that perform specific tasks intelligently, while AGI (Artificial General Intelligence) represents hypothetical AI with human-level reasoning across all domains. Current LLMs are narrow AI - powerful at language tasks but lacking true understanding or general intelligence like AGI would possess.

RAG

Retrieval-Augmented Generation (RAG) combines LLMs with external knowledge retrieval to ground responses in verified, current information. RAG retrieves relevant documents before generating responses, reducing hallucinations and enabling access to information beyond the model's training cutoff. This approach improves accuracy and provides source attribution.